claude-windows / 50e27c80-8f59-4d8f-8a14-fdbddb9fbb3a

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\50e27c80-8f59-4d8f-8a14-fdbddb9fbb3a\subagents\workflows\wf_ddb0b3fd-b40\agent-a5d051e407accd2f2.jsonl`
- SHA-256: `2f54beb88d75f0351dc3cc50a80b2378a3a76566a5f915102e2469cceb8f6ff0`
- Source modified: `2026-07-07T04:17:00+00:00`
- Imported at: `2026-07-08T16:00:26+00:00`
- project: `wf_ddb0b3fd-b40`
- session_id: `50e27c80-8f59-4d8f-8a14-fdbddb9fbb3a`

Transcript

1. user (2026-07-07T04:14:53.573Z)

```
You are reviewing an UNCOMMITTED working diff implementing issue #506 (GPU-resident WASB
inference) in the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer.
The full diff is at C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-
highlight-indexer/50e27c80-8f59-4d8f-8a14-fdbddb9fbb3a/scratchpad/506_diff.patch (read it). The
new test file tests/test_wasb_nvdec_resident.py is untracked (not in the diff) ? read it
directly from the repo.

Ground truth you must compare against:
- The VALIDATED reference implementation: C:/Users/avidu/Projects/khelsutra-guru/badminton-
highlight-indexer/scratch/gpu_resident/exp_t2_full.py (the GpuFrontend class + its grid build
ARE what was validated on a real L4: F1 0.574 ? baseline 0.582 on the hard clip). The production
port must be semantically faithful to it.
- WASB-SBDT interfaces (not in this repo; trust these facts):
TracknetV2Detector.run_tensor(imgs, affine_mats) moves imgs .to('cuda') itself and calls
postprocessor.run(preds, affine_mats) which does affine_mats[scale].cpu().numpy() then indexes
[i] per batch item; ImageDataset (eval) returns (imgs_t, hms_t, trans_outputs_inv, xys, visis,
img_paths) where trans_outputs_inv is {scale: np.float64 [2,3]}; per-frame preprocessing is PIL
RGB -> cv2.warpAffine(trans_input, (512,288), INTER_LINEAR) -> ToTensor -> Normalize(ImageNet);
get_transform uses c=(w/2,h/2), s=max(h,w); OnlineTracker uses np.Inf; WASB detector.py calls
torch.load(model_path) with no weights_only arg.
- The CPU decode path (run(), run_video_streaming) must be BEHAVIOR-UNCHANGED on the WSL
torch-1.11 + numpy<1.24 env (byte-parity guarantee), except that torch>=2.6 envs get the
weights_only + np.Inf shims.

Report ONLY defects you can ground in the actual code you read ? cite file:line. No style nits,
no speculative maybes without a concrete failure scenario. Return findings as structured output.

A reviewer claims this defect in the #506 diff:
- file: deploy/digitalocean/gpu_setup.sh
- line: 44
- summary: `sh /tmp/drv.run --extract-only --target /tmp/drv-extract` fails if /tmp/drv-extract
already exists (the NVIDIA .run extractor refuses an existing target directory), and the cleanup
`rm -rf /tmp/drv-extract /tmp/drv.run` at line 48 only executes on full success ? under `set
-euo pipefail`, any failure between extraction and cleanup (e.g. `sudo cp` needing a password in
a non-interactive session, or a sudo permission error) aborts the script and leaves the stale
extract dir behind.
- failure scenario: First run fails at `sudo cp` (non-interactive sudo prompt); the operator
fixes sudo and re-runs gpu_setup.sh; the script now fails at the extraction step with 'target
directory already exists' on every subsequent run until someone manually removes /tmp/drv-
extract ? a confusing loop on a paid GPU box that the script's own idempotent design (the if-
guard on the .so.1 files) was meant to avoid.
- severity: minor

Adversarially VERIFY it: read the actual code (and the validated reference / WASB interface
```

facts above) and try to REFUTE the claim. It is only real if the concrete failure scenario actually happens in the code as written. Default to isReal=false if the scenario cannot occur, is already guarded, or rests on a misreading.

2. user (2026-07-07T04:15:00.599Z)

```
bootstrap.sh
gpu_setup.sh
mount_scratch.sh
README.md
cloudrun
digitalocean
gcp
```

3. user (2026-07-07T04:15:02.903Z)

```
403:+                f"/ gpu_setup.sh stage them). Fall back to decode_backend=cpu."
687: # native GPU box, deploy/digitalocean/gpu_setup.sh):
711:+    && curl -fsSL -o /tmp/drv.run \
713:+    && sh /tmp/drv.run --extract-only --target /tmp/drv \
716:+    && rm -rf /tmp/drv /tmp/drv.run
765:diff --git a/deploy/digitalocean/gpu_setup.sh b/deploy/digitalocean/gpu_setup.sh
767:--- a/deploy/digitalocean/gpu_setup.sh
768:+++ b/deploy/digitalocean/gpu_setup.sh
806:+  curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/tesla/$DRV/NVIDIA-Linux-x86_64-$DRV.run" \
807:+    || curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/XFree86/Linux-x86_64/$DRV/NVIDIA-Linux-x86_64-$DRV.run"
808:+  sh /tmp/drv.run --extract-only --target /tmp/drv-extract >/dev/null
809:+  sudo cp -a /tmp/drv-extract/libnvcuvid.so.* /tmp/drv-extract/libnvidia-encode.so.* /usr/lib/x86_64-linux-gnu/
814:+  rm -rf /tmp/drv-extract /tmp/drv.run
```

4. user (2026-07-07T04:15:11.557Z)

```
760     -        hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml matplotlib "numpy<1.24"
761     +        hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml matplotlib numpy
762
763      # --- Main application env ---
764      WORKDIR /app
765     diff --git a/deploy/digitalocean/gpu_setup.sh b/deploy/digitalocean/gpu_setup.sh
766     index 8d47e49..a89cc1b 100755
767     --- a/deploy/digitalocean/gpu_setup.sh
768     +++ b/deploy/digitalocean/gpu_setup.sh
769     @@ -3,8 +3,14 @@
770      # Native (no-WSL) WASB shuttle-detector setup for a Linux GPU box ? the M3.5 enabler for
771      # real trajectory extraction. Replicates docs/TRACKNET_WSL_SETUP.md's WSL conda env
772      # *natively* on Linux. Run AFTER ./bootstrap.sh --gpu (which provides CUDA torch for the
773     -# main venv); this builds a SEPARATE conda env for WASB (torch 1.11) ? never co-installed
774     -# with the indexer's torch (framework conflict).
775     +# main venv); this builds a SEPARATE conda env for WASB ? never co-installed with the
776     +# indexer's torch (framework conflict).
777     +#
778     +# #506 modernized stack (validated end-to-end on a real L4, 2026-07-06): py3.10 +
779     +# torch 2.6.0+cu124 (WASB HRNet ports clean; cpu F1 ? the torch-1.11 baseline) +
780     +# PyNvVideoCodec for the GPU-resident decode backend (indexing.wasb.decode_backend=
781     +# nvdec_resident, ~1.7×). Step [2/6] stages the NVDEC/NVENC driver userspace libs
782     +# PyNvVideoCodec needs (compute-only driver images ship neither).
783      #
```

```
 # Requires an NVIDIA GPU + driver (the DO `gpu-*-base` image ships it).
 #
@@ -18,7 +24,7 @@ CONDA_DIR="${CONDA_DIR:-$HOME/miniconda3}"
 WASB_REPO="$MODELS_DIR/WASB-SBDT"
 mkdir -p "$MODELS_DIR"

-echo "[gpu] [1/5] miniconda (native Linux)"
+echo "[gpu] [1/6] miniconda (native Linux)"
 if [ ! -x "$CONDA_DIR/bin/conda" ]; then
   curl -sSL https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -o /tmp/mc.sh
   bash /tmp/mc.sh -b -p "$CONDA_DIR" && rm -f /tmp/mc.sh
@@ -26,29 +32,49 @@ fi
 # shellcheck disable=SC1091
 source "$CONDA_DIR/etc/profile.d/conda.sh"

+echo "[gpu] [2/6] NVDEC/NVENC driver userspace libs (#506: PyNvVideoCodec needs BOTH)"
+# Compute-only driver images (GCE DLVM, DO gpu-*-base) ship neither libnvcuvid.so.1 nor
+# libnvidia-encode.so.1. Extract them from the .run matching the RUNNING driver so the
+# userspace/kernel versions can never skew (see the #506 / gpu-vm-setup-gotchas notes).
+if [ ! -f /usr/lib/x86_64-linux-gnu/libnvcuvid.so.1 ] || [ ! -f /usr/lib/x86_64-linux-gnu/libnvidia-encode.so.1 ]; then
+  DRV="$(nvidia-smi --query-gpu=driver_version --format=csv,noheader | head -1)"
+  echo "[gpu]       extracting libnvcuvid + libnvidia-encode from driver $DRV ..."
+  curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/tesla/$DRV/NVIDIA-Linux-x86_64-$DRV.run" \
+    || curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/XFree86/Linux-x86_64/$DRV/NVIDIA-Linux-x86_64-$DRV.run"
+  sh /tmp/drv.run --extract-only --target /tmp/drv-extract >/dev/null
+  sudo cp -a /tmp/drv-extract/libnvcuvid.so.* /tmp/drv-extract/libnvidia-encode.so.* /usr/lib/x86_64-linux-gnu/
+  ( cd /usr/lib/x86_64-linux-gnu \
+    && sudo ln -sf "libnvcuvid.so.$DRV" libnvcuvid.so.1 \
+    && sudo ln -sf "libnvidia-encode.so.$DRV" libnvidia-encode.so.1 \
+    && sudo ldconfig )
+  rm -rf /tmp/drv-extract /tmp/drv.run
+else
+  echo "[gpu]       already present"
+fi
+
 # Modern conda BLOCKS non-interactive `conda create` until the defaults-channel Terms of Service
 # are accepted (CondaToSNonInteractiveError). Accept them so the env build doesn't fail on a fresh
 # box. (Observed on a GCP Deep-Learning-VM image, 2026-06-20.)
 conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main >/dev/null 2>&1 || true
 conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r >/dev/null 2>&1 || true

-echo "[gpu] [2/5] WASB-SBDT repo"
+echo "[gpu] [3/6] WASB-SBDT repo"
 if [ ! -d "$WASB_REPO/.git" ]; then
   git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO"
 fi

-echo "[gpu] [3/5] conda env 'wasb' (py3.8 + torch 1.11.0+cu113 ? the verified WASB stack)"
-conda env list | grep -q '/envs/wasb$' || conda create -y -q -n wasb python=3.8
+echo "[gpu] [4/6] conda env 'wasb' (py3.10 + torch 2.6.0+cu124 + PyNvVideoCodec ? #506 stack)"
+conda env list | grep -q '/envs/wasb$' || conda create -y -q -n wasb python=3.10
 conda activate wasb
-python -m pip install -q torch==1.11.0+cu113 torchvision==0.12.0+cu113 \
-  --extra-index-url https://download.pytorch.org/whl/cu113
```

```
-[ -f "$WASB_REPO/requirements.txt" ] && python -m pip install -q -r "$WASB_REPO/requirements.txt" || true
-# WASB-SBDT's requirements.txt is INCOMPLETE: src/ imports hydra, pandas, etc. that it omits, so
```

5. user (2026-07-07T04:15:12.338Z)

```
#!/usr/bin/env bash
#
# Native (no-WSL) WASB shuttle-detector setup for a Linux GPU box ? the M3.5 enabler for
# real trajectory extraction. Replicates docs/TRACKNET_WSL_SETUP.md's WSL conda env
# *natively* on Linux. Run AFTER ./bootstrap.sh --gpu (which provides CUDA torch for the
# main venv); this builds a SEPARATE conda env for WASB ? never co-installed with the
# indexer's torch (framework conflict).
#
# #506 modernized stack (validated end-to-end on a real L4, 2026-07-06): py3.10 +
# torch 2.6.0+cu124 (WASB HRNet ports clean; cpu F1 ? the torch-1.11 baseline) +
# PyNvVideoCodec for the GPU-resident decode backend (indexing.wasb.decode_backend=
# nvdec_resident, ~1.7×). Step [2/6] stages the NVDEC/NVENC driver userspace libs
# PyNvVideoCodec needs (compute-only driver images ship neither).
#
# Requires an NVIDIA GPU + driver (the DO `gpu-*-base` image ships it).
#
# ? WASB-C3: the pretrained badminton weights' license / source-dataset provenance is NOT
# verified here. Fine for internal extraction + Gen-0 bootstrapping; clear it before sharing
# any derived trajectories or model. See docs/DECOUPLED_COMPUTE/06.
set -euo pipefail

MODELS_DIR="${RALLY_MODELS_DIR:-$HOME/models}"
CONDA_DIR="${CONDA_DIR:-$HOME/miniconda3}"
WASB_REPO="$MODELS_DIR/WASB-SBDT"
mkdir -p "$MODELS_DIR"

echo "[gpu] [1/6] miniconda (native Linux)"
if [ ! -x "$CONDA_DIR/bin/conda" ]; then
  curl -sSL https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -o /tmp/mc.sh
  bash /tmp/mc.sh -b -p "$CONDA_DIR" && rm -f /tmp/mc.sh
fi
# shellcheck disable=SC1091
source "$CONDA_DIR/etc/profile.d/conda.sh"

echo "[gpu] [2/6] NVDEC/NVENC driver userspace libs (#506: PyNvVideoCodec needs BOTH)"
# Compute-only driver images (GCE DLVM, DO gpu-*-base) ship neither libnvcuvid.so.1 nor
# libnvidia-encode.so.1. Extract them from the .run matching the RUNNING driver so the
# userspace/kernel versions can never skew (see the #506 / gpu-vm-setup-gotchas notes).
if [ ! -f /usr/lib/x86_64-linux-gnu/libnvcuvid.so.1 ] || [ ! -f /usr/lib/x86_64-linux-gnu/libnvidia-encode.so.1 ]; then
  DRV="$(nvidia-smi --query-gpu=driver_version --format=csv,noheader | head -1)"
  echo "[gpu]       extracting libnvcuvid + libnvidia-encode from driver $DRV ..."
  curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/tesla/$DRV/NVIDIA-Linux-x86_64-$DRV.run" \
    || curl -fsSL -o /tmp/drv.run "https://us.download.nvidia.com/XFree86/Linux-x86_64/$DRV/NVIDIA-Linux-x86_64-$DRV.run"
  sh /tmp/drv.run --extract-only --target /tmp/drv-extract >/dev/null
  sudo cp -a /tmp/drv-extract/libnvcuvid.so.* /tmp/drv-extract/libnvidia-encode.so.* /usr/lib/x86_64-linux-gnu/
  ( cd /usr/lib/x86_64-linux-gnu \
    && sudo ln -sf "libnvcuvid.so.$DRV" libnvcuvid.so.1 \
    && sudo ln -sf "libnvidia-encode.so.$DRV" libnvidia-encode.so.1 \
    && sudo ldconfig )
  rm -rf /tmp/drv-extract /tmp/drv.run
else
```

```
  echo "[gpu]       already present"
fi

# Modern conda BLOCKS non-interactive `conda create` until the defaults-channel Terms of Service
# are accepted (CondaToSNonInteractiveError). Accept them so the env build doesn't fail on a fresh
# box. (Observed on a GCP Deep-Learning-VM image, 2026-06-20.)
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main >/dev/null 2>&1 || true
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r >/dev/null 2>&1 || true

echo "[gpu] [3/6] WASB-SBDT repo"
if [ ! -d "$WASB_REPO/.git" ]; then
  git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO"
fi

echo "[gpu] [4/6] conda env 'wasb' (py3.10 + torch 2.6.0+cu124 + PyNvVideoCodec ? #506 stack)"
conda env list | grep -q '/envs/wasb$' || conda create -y -q -n wasb python=3.10
conda activate wasb
python -m pip install -q torch==2.6.0 torchvision==0.21.0 \
  --index-url https://download.pytorch.org/whl/cu124
python -m pip install -q PyNvVideoCodec
# WASB-SBDT ships NO requirements.txt: src/ imports hydra, pandas, etc. ? this explicit list is
# the L4-validated dependency closure (#506). numpy is UNPINNED (the old "numpy<1.24" was the
# torch-1.11 ABI pin); wasb_infer.py shims the NumPy-2.0 np.Inf removal for WASB's tracker.
python -m pip install -q hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml matplotlib numpy || true

echo "[gpu] [5/6] pretrained badminton weights"
WEIGHTS="$WASB_REPO/pretrained_weights/wasb_badminton_best.pth.tar"
if [ -f "$WEIGHTS" ]; then
  echo "[gpu]       found $WEIGHTS"
else
  echo "[gpu]       !! MISSING $WEIGHTS ? fetch wasb_badminton_best.pth.tar (Drive/bucket) before extraction."
fi

echo "[gpu] [6/6] GPU sanity check (in the wasb env)"
python - <<'PY'
import torch
ok = torch.cuda.is_available()
print("[gpu]       torch", torch.__version__, "| cuda_available", ok,
      "| device", (torch.cuda.get_device_name(0) if ok else "CPU"))
assert ok, "CUDA not available in the wasb env ? check the GPU driver / base image"
try:
    import PyNvVideoCodec as pnvc
    print("[gpu]       PyNvVideoCodec", getattr(pnvc, "__version__", "?"), "OK")
except Exception as e:  # nvdec_resident unavailable; the cpu decode path still works
    print("[gpu]       !! PyNvVideoCodec import failed:", repr(e)[:150])
PY

echo "[gpu] native WASB env ready.  Activate with:  source $CONDA_DIR/etc/profile.d/conda.sh && conda activate wasb"
echo "[gpu] (the LinuxNativeRunner ? backend/pipeline/detectors ? drives extraction once it lands; M3.5.)"
```

6. user (2026-07-07T04:15:56.412Z)

```
/usr/bin/bash: line 1: strings: command not found
```

7. user (2026-07-07T04:16:03.024Z)

```
692:    echo "The directory '$tmpdir' already exists.  Please either"
```

8. user (2026-07-07T04:16:10.163Z)

```
[ "$LOCAL_ARCH" = "i386" ] && LOCAL_ARCH="x86"
[ "$LOCAL_ARCH" = "i486" ] && LOCAL_ARCH="x86"
[ "$LOCAL_ARCH" = "i586" ] && LOCAL_ARCH="x86"
[ "$LOCAL_ARCH" = "i686" ] && LOCAL_ARCH="x86"

if [ -z "$LOCAL_OS" -o -z "$LOCAL_ARCH" ]; then
    echo "ERROR: missing/broken uname.  Cannot perform platform check."
    exit 1;
fi

if [ "$LOCAL_ARCH" != "$TARGET_ARCH" -o "$LOCAL_OS" != "$TARGET_OS" ]; then
    if [ "$run_script" = "y" ]; then
        echo ""
        echo "ERROR: this .run file is intended for the"
        echo "${TARGET_OS}-${TARGET_ARCH} platform, but you appear to be"
        echo "running on ${LOCAL_OS}-${LOCAL_ARCH}.  Aborting installation."
        echo ""
        exit 1;
    fi
fi


if [ "$keep" = "y" ]; then
    echo "Creating directory $targetdir"; tmpdir=$targetdir;
else
    workingdir="$TMPROOT/selfgz$$"
    tmpdir="$workingdir/$targetdir";
    rm -rf $tmpdir
fi

if [ -d "$tmpdir" -o -f "$tmpdir" ]; then
    echo "The directory '$tmpdir' already exists.  Please either"
    echo "move the existing directory out of the way, or specify a"
    echo "different directory with the '--target' option."
    exit 1
fi

mkdir -p $tmpdir || {
    echo "Unable to create the target directory '$tmpdir'."
    exit 1
}

echo=echo; [ -x /usr/ucb/echo ] && echo=/usr/ucb/echo
if [ x$SETUP_NOCHECK != x1 ]; then
    $echo -n "Verifying archive integrity... "
    sum1=`tail -n +4 $0 | cksum | awk '{print $1}'`
    [ $sum1 != $CRCsum ] && {
        $echo "Error in check sums $sum1 $CRCsum"
        exit 2;
    }
    echo "OK"
fi
if [ $MD5 != \"00000000000000000000000000000000\" ]; then
    # space separated list of directories
    [ x"$GUESS_MD5_PATH" = "x" ] && GUESS_MD5_PATH="/usr/local/ssl/bin /usr/local/bin /usr/bin
/bin"
    MD5_PATH=""
```

```
    for a in $GUESS_MD5_PATH; do
        if [ -x "$a/md5sum" ]; then
            MD5_PATH=$a;
        fi
```

9. user (2026-07-07T04:16:24.996Z)

```
11:targetdir=NVIDIA-Linux-x86_64-550.90.07
13:keep=n
52:    echo "-x, --extract-only"
96:    echo "-x, --extract-only"
119:    echo "--target [NewDirectory]"
455:if [ "$1" = "--target-os" ]; then
460:if [ "$1" = "--target-arch" ]; then
465:if [ "$1" = "--target-directory" ]; then
538:keep=n
544:       "--extract-only"|"-x")
546:           keep=y;
549:           keep=y;
551:       "--target")
553:               targetdir="$2";
554:               keep=y;
557:               echo "ERROR: --target: no target directory specified."
694:    echo "different directory with the '--target' option."
782:        new_targetdir="`basename $targetdir | sed -e \"s/-custom//\"`"
783:        new_targetdir="${new_targetdir}-custom"
809:        sh ./$new_targetdir/makeself.sh              --lsm $tmplsm              --version-
string $version_string              --pkg-version $pkg_version              --pkg-history
./$new_targetdir/pkg-history.txt              --target-os $TARGET_OS              --target-arch
$TARGET_ARCH              $new_targetdir $new_targetdir.run              "$label" "$script"
```

10. user (2026-07-07T04:17:00.574Z)

```
Structured output provided successfully
```